

Symmetry-Quotient Forgery Attacks on Odd-Characteristic SNOVA

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Abstract

We give a forgery attack on the odd-characteristic form of SNOVA, a third-round candidate in NIST’s Additional Digital Signatures process. SNOVA’s original characteristic-two parameters have already been broken, and the team’s public repair moves to odd characteristic, where the public-key matrices become symmetric; we show that this repair also falls below its targeted security. In SNOVA’s own cost model, our estimates fall to between $2^{130.3}$ and $2^{138.9}$ Boolean gates at Level I, about $2^{184.3}$ at Level III, and between $2^{228.0}$ and $2^{238.8}$ at Level V, against targets 2^{143} , 2^{207} , and 2^{272} . All nine Version 2.3 draft parameter sets fall short, by 4.06 to 44.05 bits, as do the six Version 2.4 analysis-preview shapes, by as little as 1.43 bits if they keep the symmetric construction.

The attack builds on the existing Beullens forgery framework (Eurocrypt 2025), which recasts forgery as solving a system of quadratic equations whose hardness is governed by the rank of an associated linear map. We show that the new symmetry forces that map to depend only on unordered power pairs. Its universal rank ceiling is therefore $m_1 \binom{\ell+1}{2}$ rather than $m_1 \ell^2$ (here ℓ is the block size and m_1 a scheme parameter). Across the nine Version 2.3 sets, the resulting residual systems range from 48 equations in 112 variables to 96 equations in 224 variables.

The symmetric-square factorization and rank ceiling are exact for every key with symmetric public matrices. The residual-system dimensions we tabulate additionally assume full quotient rank $\rho = K$ and full affine-constraint rank $\lambda_R = M - K$; we verify these end-to-end on the audited (28, 5, 4, 4) Level-I shape and state them as explicit, checkable genericity assumptions for the other shapes. We also reconstruct the audited Level-I reduced system from the official reference implementation’s own key material.

All bit-security figures inherit the semi-regular and sparse-linear-algebra assumptions used in the SNOVA team’s analysis; we do not claim a full-size forgery. This exact quotient does not apply to the unsymmetrized $q = 16$ public matrices.

1 Introduction

SNOVA is a structured Oil-and-Vinegar signature scheme designed to combine very small public keys with short signatures and fast verification. NIST advanced SNOVA to the third round of the Additional Digital Signatures process in May 2026, while allowing the surviving candidates to submit further specifications and implementations [12, 11]. The scheme has not, however, had a single stable public parameter generation, and no definitive Round 3 parameter package was public as of 24 July 2026.

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The NIST-posted Round 2 package uses $\mathbb{F}_{16}$ and nonsymmetrized public base matrices. That line has already been heavily damaged: Beullens reduced the security margins by exploiting the structured whipping layer [1], and the subsequent structure-aware wedge attack of Bros et al. reports that all Round 2 parameters are broken, while noting a remaining rank conjecture in its complexity analysis [2]. The SNOVA team’s later public drafts pursue odd-characteristic variants of the scheme intended to evade these attacks. Version 2.1 added prime-field variants with symmetric public matrices; Version 2.3 supplies $q = 19$ implementations and KATs; the public analysis repository later posted a six-set “Version 2.4” preview; and the authors’ reformulation paper presents odd characteristic as a principal route for redesigning the scheme [16, 14, 4]. An odd-characteristic attack therefore targets the public repair strategy rather than the already-vulnerable submitted parameters.

The Version 2.3 attack chapter retains a forgery analysis in the style of Beullens [1]. There, forgery is recast as solving a system of quadratic equations, and the size of that system is taken to be $m_1 \ell^2$: the analysis lets the powers $S^0, \dots, S^{\ell-1}$ of the public matrix S contribute all ℓ^2 ordered pairs (a, b) , and it models the resulting equations as essentially random of that size. The draft itself states that this chapter is inherited from Version 2.0 and still needs updating [16].

This paper shows that the ordered model is incompatible with the new public symmetry. Put $A = \mathbb{F}_q[S]$ and $d = \dim_{\mathbb{F}_q} A$; the current irreducible degree- ℓ construction has $d = \ell$ and $A \cong \mathbb{F}_{q^\ell}$. Since $S = S^\top$, every element of A is symmetric. For each public base form P_i , exchanging the two labels in

$$(\xi, \eta) \longmapsto u^\top (I_n \otimes \xi) P_i (I_n \otimes \eta) u$$

does not change the resulting quadratic polynomial in u . Equivalently, the ordered label map $A \otimes_{\mathbb{F}_q} A$ annihilates every tensor $\xi \otimes \eta - \eta \otimes \xi$ and factors through the quotient $\text{Sym}_{\mathbb{F}_q}^2(A)$. In odd characteristic the alternating tensors form a canonical $\binom{d}{2}$ -dimensional subspace of this kernel, giving

$$A \otimes_{\mathbb{F}_q} A \cong \text{Sym}_{\mathbb{F}_q}^2(A) \oplus \bigwedge_{\mathbb{F}_q}^2(A).$$

The alternating subspace is a universal kernel contribution; additional key- or relation-dependent kernel may exist. Thus the universal homogeneous-rank ceiling is

$$K_d = m_1 \binom{d+1}{2},$$

and for the current parameters this is $K = m_1 \binom{\ell+1}{2}$ rather than $m_1 \ell^2$. The cap holds for every key generated with symmetric public matrices; it is not a rare MinRank event and cannot be removed by adding more A, B, Q terms (the “emulsifier” matrices of the whipping layer, which randomize the public map).

1.1 Results

Recall the parameters: v vinegar and o oil matrix variables (so $n = v + o$), block size ℓ , and r columns per signature block, with $m_1 = \lceil or/\ell \rceil$ scalar-expanded public base matrices and $M = or\ell$ scalar verification equations. Our main algebraic result is the following.

Theorem 1 (Universal symmetry quotient, informal). *Consider a SNOVA public key for which the power matrix S and the scalar-expanded public base matrices $P_1, \dots, P_{m_1}$ are symmetric, and write $d = \dim_{\mathbb{F}_q} \mathbb{F}_q[S]$. After any affine-column substitution in the Beullens framework, the homogeneous quadratic part factors through*

$$\bigoplus_{i=1}^{m_1} \text{Sym}_{\mathbb{F}_q}^2(\mathbb{F}_q[S])$$

and has rank at most $K_d = m_1 \binom{d+1}{2}$. For the current irreducible degree- ℓ construction, $d = \ell$ and $K_d = K = m_1 \binom{\ell+1}{2}$.

For a fixed relation and offsets, let $\rho \leq K$ be the actual quotient rank and let λ_R be the rank of the induced affine constraints. Whenever those constraints are consistent with the target, forging is exactly reducible to ρ equations of degree at most two in $\ell(v + o) - \lambda_R$ variables. In the full-rank case $\rho = K$ and $\lambda_R = M - K$, every target is consistent and the residual dimensions are

$$K \text{ equations in } N_{\text{res}} = \ell(v + o) - (o\ell - K) \text{ variables.}$$

The full theorem, including nongeneric ranks, appears as [Theorems 4 and 6](#). The rank cap comes from the domain of the quadratic feature map. The A, B, Q emulsifier matrices may randomize the induced map on the symmetric quotient, but they cannot recover the alternating coordinates discarded before emulsification. For a simple relation, we certify full quotient rank for all nine Version 2.3 shapes. For the rejection-bypassing relation used in the end-to-end attack, we certify both quotient and affine ranks on the audited (28, 5, 4, 4) Level-I KAT-derived key and two further generated keys of the same shape. For every other shape, the simultaneous rank conditions for a rejection-bypassing relation are stated explicitly as assumptions in the cost tables ([Section 9](#)).

Scope of exactness. The result has three layers with different status. The symmetric-square factorization and rank ceiling are universal algebraic statements. The displayed selected-target K -in- N_{res} systems require the instance-checkable rank equalities $\rho = K$ and $\lambda_R = M - K$. Converting those dimensions into gate exponents additionally uses the submission’s heuristic semi-regular solver model. The tables below state the latter two conditions rather than folding them into the universal theorem.

Applying the submission’s own MQ methodology gives the Version 2.3 estimates in [Table 1](#). The submission’s estimator admits two conventions for the semi-regular series, differing by whether one homogenizing variable is inserted. The audited analysis snapshot and the values printed in Version 2.3 Table 11 use the convention without that variable ($p_1 = 0$); inserting the extra variable ($p_1 = 1$) yields a stricter, higher estimate. We adopt the stricter $p_1 = 1$ convention for the headline figures, so our reported costs are conservative relative to the submission’s own numbers; even so, every $q = 19$ set falls below target. Applying the identical quotient to the Version 2.4 analysis preview gives the additional estimates in [Table 4](#).

SNOVA’s verifier rejects any signature whose blocks are symmetric. Because the attack is built from symmetric column relations, one might hope this rule blocks it. We show it does not: in the square variants the attacker adds fixed offsets that force a nonzero antisymmetric entry into every block, so the forged signature passes the check while the system to be solved is unchanged, and in the rectangular variants the blocks are not symmetric objects in the first place.

We also report an implementation audit against the reference code. Reconstructing the official Level-I implementation reproduces the reduced system exactly from its shipped key material. Independently, after fixing one signature column whose cross-term coefficient map is surjective and choosing a target-matching coset of its kernel, the verifier restricts to a sound system of 50 equations

Table 1: Symmetry-reduced residual systems and estimated attack costs in the full-rank case $\rho = K$, $\lambda_R = M - K$. A simple relation has certified full quotient rank on all nine shapes. Simultaneous quotient and affine rank for the rejection-bypassing attack relation is certified end-to-end only on the audited (28, 5, 4, 4) Level-I shape and is an explicit genericity assumption for the other rows. “Old” is the Beullens-forgery estimate in SNOVA Version 2.3 Table 11. “New” uses the stricter $p_1 = 1$ convention. Negative margins mean that the estimate is below the category target.

Level	Parameters (v, o, ℓ, r)	Residual MQ	Old	New	Target	Margin
I	(28, 5, 4, 4)	50 in 102	194	138.94	143	−4.06
I	(48, 16, 2, 2)	48 in 112	164	130.30	143	−12.70
I	(28, 4, 4, 5)	50 in 98	194	138.94	143	−4.06
III	(40, 7, 4, 4)	70 in 146	264	184.29	207	−22.71
III	(72, 24, 2, 2)	72 in 168	234	184.29	207	−22.71
III	(38, 5, 4, 5)	70 in 142	239	184.29	207	−22.71
V	(50, 9, 4, 4)	90 in 182	333	227.95	272	−44.05
V	(96, 32, 2, 2)	96 in 224	303	238.77	272	−33.23
V	(52, 6, 4, 6)	90 in 178	333	227.95	272	−44.05

of degree at most two in 52 variables (Sections 7 and 9). In the reported structural tests, no shared solution locus or exploitable low-rank shortcut was observed beyond this reduction; this is negative evidence rather than a completeness result.

1.2 Organization

Section 2 surveys prior work. Section 3 fixes notation and recalls the affine-column framework. Section 4 proves the quotient and its multibase Gram extension. Section 5 derives the smaller system of equations a forger must solve, and Section 6 shows that the verifier’s rule against symmetric signatures does not block the attack. Section 7 gives the cross-column 50-in-52 normal form. Section 8 evaluates our attack’s runtime against both the Version 2.3 parameter sets and the Version 2.4 preview. Section 9 reports the experiments confirming the reduction against SNOVA’s reference implementation and testing its solving-degree model. Section 10 gives countermeasures. Appendix C rules out an obvious alternative attack—a MinRank shortcut over the extension field—and Appendix D develops the multibase solver extension.

2 Related Work

Beullens’s forgery attack. Beullens identified SNOVA’s structured whipping layer, expressed the public map through ordered bilinear power-pair coordinates, and showed that low-rank combinations of the corresponding emulsifier matrices lead to forgery attacks [1]. The Version 2.3 countermeasure increases the number and diversity of A, B, Q terms to make the induced ordered-coordinate matrix resemble a random matrix.

Our attack uses the same affine-column architecture but a different source of rank loss. We do not find a low-rank element in a family of otherwise full-rank ordered-coordinate matrices. We show that the odd-characteristic homogeneous feature vector itself lies in a K -dimensional symmetric subspace. Thus the quotient applies to every key and every relation, even when the induced map is maximally random and full rank on that quotient.

Stable-ideal attacks. Cabarcas et al. exploit the action of the commutative group generated by S to design a polynomial-system solver specialized to SNOVA [3]. Their work can be viewed as improving the residual-system solving stage. The present result instead reduces the number of residual equations before solving begins. The approaches are compatible, but we do not combine their cost benefits here.

Characteristic-two wedge attacks and the odd-characteristic redesign. The structure-aware wedge attack exploits SNOVA’s block-ring structure and reports that the NIST Round 2 $q = 16$ parameters are broken, while explicitly retaining a conjectural rank component in the analysis [2]. The SNOVA authors subsequently investigated the wedge map and proposed a reformulation emphasizing odd-characteristic and symmetric-form variants [15, 4]. Their earlier Round 2 security paper explicitly noted that symmetric public matrices over odd-prime fields reduce the number of alternating forms by roughly a factor of two, cautioned that symmetrization may create new attacks, and proposed excluding symmetric signature matrices [WangEtAl2025Round2Security]. That observation was qualitative and made in the wedge-attack setting. Our attack is complementary: it does not apply to the unsymmetrized $q = 16$ public matrices, but it attacks the public odd-characteristic repair direction through an exact affine-column quotient that is independent of the wedge-rank conjecture.

Key-recovery attacks. Several works interpret the scalar expansion of SNOVA as UOV, or lift it to smaller extension-field UOV instances, and transfer key-recovery attacks [6, 9, 10]. Our result is a universal forgery reduction and does not recover the secret oil space.

Relation to the symmetric-algebra key-recovery attack. The decomposition $V \otimes V = \text{Sym}^2(V) \oplus \wedge^2(V)$ and the vanishing of the alternating part under quadratic evaluation are classical; we claim no novelty for that algebra [8]. The closest cryptanalytic precedent is the p -reduced symmetric-algebra framework of Jin, Pan, He, Gong, and Ding [7], which generalizes Ran’s characteristic-two wedge attack [13] to odd characteristic. Three differences are worth stating. First, theirs is a *key-recovery* XL attack that reconstructs the oil space, whereas ours is a universal forgery that recovers no secret. Second, their algebra is the quotient $\mathbb{F}_q[x_1, \dots, x_n]/(x_1^p, \dots, x_n^p)$ on all $n = v + o$ variables, in which the polar form is a degree-two element and the Macaulay computation runs to degree up to $o(p - 1)$; our symmetric square is the degree-two part of the ℓ -dimensional power algebra $\mathbb{F}_q[S]$, and collapses only the ordered power pairs of the Beullens feature map. Third, their cost is strongly characteristic-sensitive—it grows with p and is ineffective at the large characteristics of QR-UOV—whereas our quotient dimension $m_1^{\ell+1}$ is independent of p for odd p . The two attacks are therefore complementary, and ours applies directly in the moderate-characteristic regime $q = 19$ that the odd-characteristic SNOVA drafts adopt. Their own concrete recommended-parameter evaluation targets QR-UOV and reports no improvement over the existing reconciliation attack at its large characteristic; they provide no SNOVA cost tables, so no numerical head-to-head on the $q = 19$ Version 2.3 sets is available.

Our contribution. The contribution is to turn the prior qualitative symmetry warning into the exact $\text{Sym}^2(A)$ factorization of the Beullens affine-column feature map; derive the rank and selected-target residual-system consequences for square and rectangular variants; extend the factorization to the multibase Gram setting; prove constructively that the proposed signature rejection does not block the attack; and validate the reduction against the audited Version 2.3 implementation snapshot and cost methodology. We found no exact affine-column quotient, selected-target reduction,

or revised cost table in the Version 2.3 attack chapter or the Version 2.4 analysis preview. The former continues to use ordered ℓ^2 coordinates and explicitly states that it still requires updating.

3 Preliminaries and the Existing Forgery Framework

3.1 Parameters and scalar expansion

Let q be a prime power, let ℓ be the size of the square matrix blocks, and let r be the number of columns of each signature block. The public map has $n = v + o$ matrix variables. Version 2.3 defines

$$m_1 = \left\lceil \frac{or}{\ell} \right\rceil \quad \text{and} \quad M = or\ell,$$

where m_1 is the number of scalar-expanded public base matrices and M is the number of scalar output equations. A signature is an n -tuple of blocks in $\text{Mat}_{\ell \times r}(\mathbb{F}_q)$. Equivalently, writing its columns as

$$U = [u_0 \mid u_1 \mid \cdots \mid u_{r-1}], \quad u_j \in \mathbb{F}_q^{n\ell},$$

the attack starts with $n\ell$ variables after expressing the other columns affinely in terms of u_0 .

Let $S \in \text{Mat}_\ell(\mathbb{F}_q)$ be the public symmetric matrix with irreducible characteristic polynomial, and put

$$\Sigma_a = I_n \otimes S^a \in \text{Mat}_{n\ell}(\mathbb{F}_q), \quad 0 \leq a < \ell.$$

For each base matrix $P_i \in \text{Mat}_{n\ell}(\mathbb{F}_q)$ define the bilinear power-pair forms

$$B_i^{(a,b)}(x, y) = x^\top \Sigma_a P_i \Sigma_b y. \quad (1)$$

The public odd-characteristic algorithms generate each scalar-expanded public matrix symmetrically: diagonal blocks are symmetric and opposite off-diagonal blocks are transposes. Thus $P_i^\top = P_i$ for the public $q = 19$ drafts, which we confirm directly against the official Level-I KAT in [Section 9](#).

3.2 The Beullens affine-column substitution

Beullens observed that the SNOVA public map can be written as a linear transformation of the values $B_i^{(a,b)}(u_j, u_k)$ [1]. The attacker selects matrices $R_j \in \mathbb{F}_q[S]$ and offsets $v_j \in \mathbb{F}_q^{n\ell}$, with $R_0 = I$, and substitutes

$$u_j = (I_n \otimes R_j)u + v_j, \quad 0 \leq j < r. \quad (2)$$

after identifying $u = u_0$ and taking $v_0 = 0$ if desired.

For a fixed relation $R = (R_0, \dots, R_{r-1})$, the substituted verification map for a target $t \in \mathbb{F}_q^M$ can be written

$$E_R z_{\text{ord}}(u) + L_R u + c_R = t. \quad (3)$$

Here

$$z_{\text{ord}}(u) = \left(B_i^{(a,b)}(u, u) \right)_{1 \leq i \leq m_1, 0 \leq a, b < \ell} \in \mathbb{F}_q^{m_1 \ell^2},$$

$E_R \in \mathbb{F}_q^{M \times m_1 \ell^2}$ depends only on public constants and the chosen column relation, while $L_R u + c_R$ collects all cross and constant terms involving the offsets.

In the ordered-coordinate analysis, if $\text{rank}(E_R) = \rho$, Gaussian elimination supplies $M - \rho$ affine-linear equations and leaves at most ρ independent quadratic equations. The Version 2.3 attack chapter models E_R as essentially random and bases its concrete estimate on ρ near the ordered-coordinate maximum. Our observation is that, whenever the public base matrices are symmetric, E_R is applied to a feature vector with deterministic coordinate equalities.

4 The Symmetry Quotient

Lemma 2 (Power-pair symmetry). *Suppose $S^\top = S$ and $P_i^\top = P_i$. Then for every $u \in \mathbb{F}_q^{n\ell}$ and every $0 \leq a, b < \ell$,*

$$B_i^{(a,b)}(u, u) = B_i^{(b,a)}(u, u).$$

Proof. The quantity is a scalar, so it equals its transpose:

$$\begin{aligned} u^\top \Sigma_a P_i \Sigma_b u &= \left(u^\top \Sigma_a P_i \Sigma_b u \right)^\top \\ &= u^\top \Sigma_b^\top P_i^\top \Sigma_a^\top u = u^\top \Sigma_b P_i \Sigma_a u. \end{aligned}$$

We used that every power of the symmetric matrix S is symmetric. $\square$

Theorem 3 (Structural symmetric-square factorization). *Let $A = \mathbb{F}_q[S] \subseteq \text{Mat}_\ell(\mathbb{F}_q)$, put $d = \dim_{\mathbb{F}_q} A$, and assume $S^\top = S$ and $P_i^\top = P_i$. Let $V = \mathbb{F}_q^{n\ell}$ and let $\mathcal{Q}_2(V)$ denote the space of formal homogeneous quadratic polynomials on V . Define the linear label map*

$$\Psi_i : A \otimes_{\mathbb{F}_q} A \longrightarrow \mathcal{Q}_2(V), \quad \Psi_i(\xi \otimes \eta)(u) = u^\top (I_n \otimes \xi) P_i (I_n \otimes \eta) u.$$

Then

$$\Psi_i(\xi \otimes \eta) = \Psi_i(\eta \otimes \xi) \quad (\xi, \eta \in A).$$

Consequently Ψ_i annihilates every alternating relation $\xi \otimes \eta - \eta \otimes \xi$ and factors uniquely through the quotient $\text{Sym}_{\mathbb{F}_q}^2(A)$. The combined label map for $P_1, \dots, P_{m_1}$ therefore factors through

$$\bigoplus_{i=1}^{m_1} \text{Sym}_{\mathbb{F}_q}^2(A),$$

a space of dimension $m_1 \binom{d+1}{2}$.

If the characteristic is odd, antisymmetrization canonically embeds $\bigwedge_{\mathbb{F}_q}^2(A)$ into $A \otimes_{\mathbb{F}_q} A$, and

$$\bigoplus_{i=1}^{m_1} \bigwedge_{\mathbb{F}_q}^2(A) \subseteq \ker \left(\bigoplus_i \Psi_i \right).$$

This is a universal kernel contribution of dimension $m_1 \binom{d}{2}$; equality with the full kernel is not asserted. If the minimal polynomial of S has degree ℓ —in particular, for the irreducible degree- ℓ SNOVA construction—then $d = \ell$.

Proof. Every element of A is a polynomial in the symmetric matrix S and is therefore symmetric; irreducibility is not needed for this step. The displayed map is linear on $A \otimes A$. For $\xi, \eta \in A$ and $u \in V$, its value is a scalar and hence equals its transpose:

$$\begin{aligned} \Psi_i(\eta \otimes \xi)(u) &= u^\top (I_n \otimes \eta) P_i (I_n \otimes \xi) u \\ &= \left(u^\top (I_n \otimes \eta) P_i (I_n \otimes \xi) u \right)^\top \\ &= u^\top (I_n \otimes \xi) P_i (I_n \otimes \eta) u = \Psi_i(\xi \otimes \eta)(u). \end{aligned}$$

Thus the alternating relations lie in the kernel, and the universal property of the symmetric-square quotient gives the factorization. Its dimension is $\binom{d+1}{2}$ per base form. When 2 is invertible, the symmetrizer and antisymmetrizer give the canonical direct decomposition $A \otimes A \cong \text{Sym}^2(A) \oplus \bigwedge^2(A)$, proving the kernel inclusion. Finally, d is the degree of the minimal polynomial of S ; an irreducible characteristic polynomial of degree ℓ therefore gives $d = \ell$. $\square$

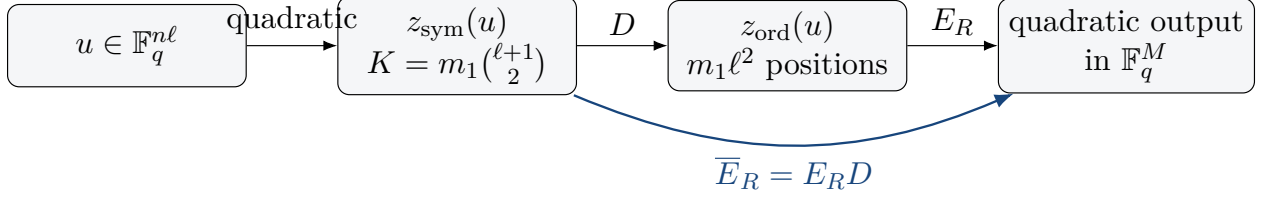


Figure 1: The ordered feature vector lies in the image of a K -dimensional duplication map. Randomizing E_R cannot increase the rank of the composite beyond K .

Concretely, for $\ell = 4$ and $d = \ell$, the ordered label space has dimension 16, while its symmetric-square quotient has dimension 10. The six alternating directions are always killed, although additional kernel may occur after the public output map is applied. With $m_1 = 5$, the universal quotient removes 30 ordered coordinates; in the audited Level-I relation the induced quotient map has full rank 50, giving the observed 30-dimensional left kernel.

Define the unordered feature vector

$$z_{\text{sym}}(u) = \left(B_i^{(a,b)}(u, u) \right)_{1 \leq i \leq m_1, 0 \leq a \leq b < \ell} \in \mathbb{F}_q^K, \quad K = m_1 \binom{\ell+1}{2}.$$

Let

$$D : \mathbb{F}_q^K \longrightarrow \mathbb{F}_q^{m_1 \ell^2}$$

be the duplication map that sends an unordered coordinate $(i, \{a, b\})$ to both ordered positions (i, a, b) and (i, b, a) when $a \neq b$, and to the single diagonal position when $a = b$. Then

$$z_{\text{ord}}(u) = D z_{\text{sym}}(u). \quad (4)$$

Theorem 4 (Symmetry rank bound). *For every public key whose power matrix and scalar-expanded public base matrices are symmetric, and for every affine-column relation R , the following holds. The homogeneous quadratic part in Equation (3) factors as*

$$\overline{E}_R z_{\text{sym}}(u), \quad \overline{E}_R = E_R D \in \mathbb{F}_q^{M \times K}.$$

In particular,

$$\text{rank}(\overline{E}_R) \leq K = m_1 \binom{\ell+1}{2}.$$

Proof. The structural factorization is Theorem 3. In the power basis $1, S, \dots, S^{\ell-1}$, it becomes the coordinate identity Equation (4). Substituting that identity into Equation (3) gives the displayed factorization, and the rank bound follows from the number of columns of $\overline{E}_R$. $\square$

Why the emulsifier does not remove the relation. The Version 2.3 changes use many independently generated $A_{i,\alpha}, B_{i,\alpha}, Q_{i,\alpha,1}, Q_{i,\alpha,2}$ matrices so that the ordered-coordinate E_R resembles a general matrix rather than the repeated block-diagonal matrix attacked in the first-round construction. This may make $\overline{E}_R$ a generic map on its K -dimensional domain. It cannot make that domain larger. The lost antisymmetric coordinates vanish before the emulsifier is applied.

The reduction is universal, not a weak-key event. Prior analyses search over relations R to find an unusually low rank. Here the cap K holds for every relation and every public key satisfying the public-matrix symmetry rule. A further quotient-rank drop below K , when accompanied by full affine rank, gives no larger residual size parameters and lowers the candidate estimator used below. Because quotient and affine ranks can interact, both should nevertheless be audited on each concrete relation.

4.1 The Gram viewpoint and several base vectors

The same argument gives a clean generalization of the attack. Choose c independent base vectors $x_1, \dots, x_c$ and an A -valued linear column relation

$$u_j = \sum_{s=1}^c (I_n \otimes R_{j,s})x_s + v_j, \quad R_{j,s} \in A. \quad (5)$$

Put $V_c = \mathbb{F}_q^c \otimes_{\mathbb{F}_q} A$, of dimension cd , where $d = \dim_{\mathbb{F}_q} A$. For each public base matrix P_i , the homogeneous restriction of the verifier is a symmetric Gram pairing on V_c . We therefore obtain the following exact extension.

Proposition 5 (*c*-base Gram factorization). *Under Equation (5), the homogeneous feature map factors through*

$$\bigoplus_{i=1}^{m_1} \text{Sym}_{\mathbb{F}_q}^2(V_c),$$

and hence through at most

$$K_{c,d} = m_1 \binom{cd+1}{2}$$

quadratic coordinates. For the current irreducible degree- ℓ construction, $d = \ell$ and this becomes $K_c = m_1 \binom{c\ell+1}{2}$. If $c = r$ and the $r \times r$ relation matrix over A is invertible, the relation is merely a public change of basis among the signature columns and imposes no restriction on the signature space.

Proof. Apply the proof of Theorem 3 to the labels $e_s \otimes \xi \in V_c$. Symmetry of the public form identifies a pair of labels with its reversal, so the universal feature space is the symmetric square of V_c . An invertible A -linear column change is bijective. $\square$

The Gram formulation also closes a broader search over arbitrary coefficient matrices in the current irreducible degree- ℓ construction. For a general relation with $R_{j,s} \in \text{Mat}_{\ell}(\mathbb{F}_q)$, define the right power-orbit space

$$\mathcal{W}_s = \text{span}_{\mathbb{F}_q} \{S^a R_{j,s} : 0 \leq a < \ell, 0 \leq j < r\}.$$

If base s is active, some $R_{j,s}$ is nonzero. The map $A \rightarrow \text{Mat}_{\ell}(\mathbb{F}_q)$, $\xi \mapsto \xi R_{j,s}$, is injective because every nonzero $\xi \in A$ is invertible. Therefore

$$\dim_{\mathbb{F}_q} \mathcal{W}_s \geq \ell. \quad (6)$$

An A -valued relation attains equality for each active base. Thus allowing arbitrary block coefficients cannot use fewer than ℓ power directions per active base; it normally makes the orbit spaces much larger. Here dimensions across different bases are counted in the labelled direct sum $\bigoplus_{s=1}^c \mathcal{W}_s$:

as unlabelled subspaces of $\text{Mat}_\ell(\mathbb{F}_q)$, distinct A -valued orbit spaces may coincide. In 500 trials per $c \in \{1, 2, 3, 4\}$ on the Level-I shape, the labelled sum $\sum_s \dim \mathcal{W}_s$ was always $c\ell$ for A -valued relations and $c\ell^2$ for unrestricted random $\ell \times \ell$ coefficients. Accidental rank defects of the public Gram map remain possible, but a broader coefficient search cannot improve the universal feature count by shrinking an active power-orbit space below the A -valued construction.

For the official Level-I constants, the quotient ranks for $c = 1, 2, 3, 4$ are respectively 50, 80, 80, 80, equal to the generic maximum $\min\{K_c, 80\}$ in every tested relation. Thus the tested $c \geq 2$ linear feature-to-output maps are surjective. This removes only the *linear* question whether a target lies in the image of that map; it does not show that the nonlinear Gram-feature map realizes a point in every target fibre. The remaining task is a structured nonlinear inversion problem. We discuss the resulting candidate multibase architecture and its present solver uncertainty in [Appendix D](#).

5 The Symmetry-Assisted Forgery Attack

Let

$$\rho = \text{rank}(\overline{E}_R) \leq K.$$

Choose a full-rank matrix $W_R \in \mathbb{F}_q^{(M-\rho) \times M}$ whose rows span the left kernel of $\overline{E}_R$. Multiplying [Equation \(3\)](#) by W_R eliminates every quadratic term:

$$W_R L_R u = W_R(t - c_R). \quad (7)$$

Let $\lambda_R = \text{rank}(W_R L_R)$. Gaussian elimination expresses λ_R coordinates of u as affine functions of the remaining

$$N_R = n\ell - \lambda_R$$

coordinates. Substitution into a row basis of [Equation \(3\)](#) leaves at most ρ quadratic equations.

Theorem 6 (Residual MQ reduction). *Fix a relation and offsets. Let $\rho = \text{rank}(\overline{E}_R)$, let W_R have rows spanning the left kernel of $\overline{E}_R$, and put $\lambda_R = \text{rank}(W_R L_R)$. Choose $V_R \in \mathbb{F}_q^{\rho \times M}$ so that*

$$T_R = \begin{bmatrix} V_R \\ W_R \end{bmatrix} \in \text{GL}_M(\mathbb{F}_q).$$

If the affine system

$$W_R L_R u = W_R(t - c_R)$$

is consistent, then one can compute a bijective affine parametrization $u = u_0 + Jx$ of its solution set, with $x \in \mathbb{F}_q^{n\ell - \lambda_R}$. Under this parametrization, solutions of the original M verification equations are in bijection with solutions of the ρ equations

$$V_R \overline{E}_R z_{\text{sym}}(u_0 + Jx) + V_R L_R(u_0 + Jx) = V_R(t - c_R),$$

each of degree at most two. Thus forging reduces in polynomial time to solving at most ρ genuinely quadratic equations in $n\ell - \lambda_R$ variables, where

$$\rho \leq m_1 \binom{\ell + 1}{2}, \quad \lambda_R \leq M - \rho.$$

In the full-rank case $\rho = K$ and $\lambda_R = M - K$, the matrix $W_R L_R$ has full row rank and the affine system is consistent for every target. The residual dimensions are therefore

$$K = m_1 \binom{\ell + 1}{2} \text{ equations of degree at most two in } N_{\text{res}} = n\ell - M + K \text{ variables.} \quad (8)$$

Every residual solution lifts by linear algebra to a signature satisfying all M original verification equations.

Proof. The rows of W_R span the complete left kernel of $\bar{E}_R$, so multiplication by the invertible matrix T_R transforms the original system into two equivalent blocks:

$$\begin{aligned} V_R \bar{E}_R z_{\text{sym}}(u) + V_R L_R u &= V_R(t - c_R), \\ W_R L_R u &= W_R(t - c_R). \end{aligned}$$

Consistency of the second block gives an affine solution space of dimension $n\ell - \lambda_R$ and a bijective parametrization $u = u_0 + Jx$. Substitution into the first block gives exactly ρ equations of degree at most two; some may lose their quadratic part after restriction, so there are at most ρ genuinely quadratic equations. Since T_R is invertible and the parametrization is bijective onto the affine solution space, this operation preserves and reflects solutions. In the full-rank case $W_R L_R$ is a surjective map onto $\mathbb{F}_q^{M-K}$, which proves target consistency and the displayed dimensions. $\square$

Proposition 7 (Effect of a rank defect). *Two departures from the full-rank case behave differently.*

1. A quotient-rank defect $\rho < K$ accompanied by full affine rank $\lambda_R = M - \rho$ leaves ρ equations of degree at most two in $n\ell - M + \rho$ variables and imposes no condition on the target. Both the equation count and the variable count are no larger than in the full-quotient case, and the estimator used in this paper assigns no larger cost. We do not claim a solver-independent monotonicity theorem for arbitrary structured MQ families.
2. An affine-rank defect $\lambda_R < M - \rho$ imposes $(M - \rho) - \lambda_R$ independent consistency conditions on the target. For a uniform target these hold with probability exactly $q^{-((M-\rho)-\lambda_R)}$. For the scheme's hash-to-field encoding this is the corresponding random-oracle approximation, so salt resampling incurs that inverse probability in expectation unless the attacker finds a relation and offsets with larger affine rank.

Proof. For (1), full row rank of $W_R L_R$ makes the affine block surjective and leaves the dimensions stated by [Theorem 6](#). The final cost comparison is a statement about the explicit estimator evaluated in this paper. For (2), the image of $W_R L_R$ is a λ_R -dimensional subspace of $\mathbb{F}_q^{M-\rho}$. Since W_R has full row rank, a uniform t induces a uniform right-hand side $W_R(t - c_R)$, which lies in that image with the stated probability. $\square$

The tabulated systems are the generic full-rank case. The residual dimensions in our tables are stated for $\rho = K$ and $\lambda_R = M - K$, where the affine subsystem has full row rank, [Equation \(7\)](#) is unconditionally solvable, and the reduction is a genuine selected-target forgery. A simple public relation attains $\rho = K$ on all nine Version 2.3 shapes and across the larger Level-I relation search. The relation equipped with the fixed skew offsets of [Lemma 8](#) is audited jointly: both equalities hold end-to-end on the KAT-derived (28, 5, 4, 4) Level-I key and two further generated keys of that same shape ([Section 9](#)). The first certificate does not imply the second, because quotient and affine ranks depend on the chosen relation and offsets. Consequently the simultaneous full-rank conditions are directly evidenced only for the audited shape and remain explicit genericity assumptions for every other tabulated shape; a defect would alter the residual dimensions or add the target-consistency cost of [Proposition 7](#).

Attack algorithm. Full affine rank removes the linear target-membership obstruction, but it does not by itself prove that every restricted nonlinear fiber contains a root. Under the same residual-solvability, guessing, and specialization assumptions used by the cost model in [Section 8](#), the reduction gives a selected-message forgery attack in the key-only setting: the attacker targets a message of its choice and makes no signing queries. If the affine block is inconsistent, or if a sampled residual fiber has no root, the procedure resamples salts or relations. For an arbitrary message $\mathcal{M}$:

1. sample a salt and compute the target $t = H(\mathcal{M}||\text{salt})$;
2. choose a public column relation R and offsets v_j ;
3. construct $\bar{E}_R, L_R, c_R$ and row-reduce as above;
4. solve the residual underdetermined MQ system;
5. reconstruct u and then the complete signature matrix U using [Equation \(2\)](#).

The first, second, fourth, and fifth steps are the same attack architecture used by Beullens. The new ingredient is that step 3 must quotient the homogeneous features by symmetry before estimating their rank.

6 Bypassing the Symmetric-Block Rejection

The SNOVA team’s 2025 security paper proposed excluding symmetric signature matrices as a mitigation for attacks that odd-characteristic symmetrization might introduce [[WangEtAl2025Round2Security](#)]. Version 2.3 implements a thresholded version of this rule: for odd q , the verifier rejects square signatures with any symmetric block when $\ell > 2$, and with more than $\lfloor n/4 \rfloor$ symmetric blocks when $\ell = 2$ [[16](#)].

The affine-column attack has fixed offsets in addition to its free vector u . Those offsets can enforce nonsymmetry identically.

Lemma 8 (A constant-skew column relation). *Assume $\ell \geq 2$ and that the characteristic polynomial of S is irreducible. Let e_0, e_1 be two distinct standard basis vectors of $\mathbb{F}_q^\ell$. There is a unique $R_1 \in \mathbb{F}_q[S]$ such that*

$$e_0^\top R_1 = e_1^\top.$$

It is computable by one $\ell \times \ell$ linear solve. With $R_0 = I$, choose offsets for the first two signature columns so that, in each matrix block,

$$(v_1)_0 - (v_0)_1 = \delta$$

for a fixed $\delta \in \mathbb{F}_q^\times$. Then every assignment to u satisfies

$$U[0, 1] - U[1, 0] = \delta$$

in that block. In particular, the block is never symmetric.

Proof. Consider the linear map

$$\theta : \mathbb{F}_q[S] \longrightarrow \mathbb{F}_q^\ell, \quad R \longmapsto e_0^\top R.$$

If $R \neq 0$, irreducibility makes R invertible, so $e_0^\top R \neq 0$. Hence θ is injective; both spaces have dimension ℓ , so it is an isomorphism. This proves existence and uniqueness. Constructively, let G be the $\ell \times \ell$ matrix whose a th row is $e_0^\top S^a$. Solve $c^\top G = e_1^\top$ and set $R_1 = \sum_{a=0}^{\ell-1} c_a S^a$.

Under the relation $u_0 = u + v_0$ and $u_1 = (I_n \otimes R_1)u + v_1$, the variable-dependent parts of the entries $U[0, 1]$ and $U[1, 0]$ are $e_0^\top R_1 u$ and $e_1^\top u$, respectively, and therefore coincide. Their difference is the chosen offset constant δ . $\square$

The construction can be applied independently to every block. It does not change the homogeneous quadratic map $\overline{E}_R$; offsets affect only L_R and c_R . For the square $\ell = 2$ sets it makes every block nonsymmetric, which is stronger than the verifier's threshold. For the square $\ell = 4$ sets it satisfies the zero-symmetric-block requirement. The rectangular variants have blocks in $\text{Mat}_{\ell \times r}$ with $r \neq \ell$, so matrix symmetry is not defined and no such rejection is applied. This bypass is algebraically universal. Whether a chosen relation and these offsets simultaneously give full affine rank is a separate, instance-checkable condition; the Level-I end-to-end audit verifies that condition for the relation used there.

7 A Cross-Column Gram Normal Form

The multibase Gram viewpoint yields a second exact reduction for the official Level-I square parameters. Write the four signature columns as $x_0, x_1, x_2, x_3 \in \mathbb{F}_{19}^{132}$. The complete symmetry-reduced feature space consists of four self blocks and six cross blocks. For a pair $s < t$, let

$$G_{st}(x_s, x_t) = (B_i^{(a,b)}(x_s, x_t))_{1 \leq i \leq 5, 0 \leq a, b < 4} \in \mathbb{F}_{19}^{80},$$

and let $E_{st} : \mathbb{F}_{19}^{80} \rightarrow \mathbb{F}_{19}^{80}$ be the corresponding public output map.

For the self contribution of column x_1 , let

$$z_1^{\text{self}}(x_1) \in \mathbb{F}_{19}^{50}, \quad E_{11}^{\text{self}} : \mathbb{F}_{19}^{50} \longrightarrow \mathbb{F}_{19}^{80}, \quad Q_1(x_1) = E_{11}^{\text{self}} z_1^{\text{self}}(x_1).$$

Here z_1^{self} is quadratic, whereas E_{11}^{self} is linear.

Proposition 9 (Official Level-I cross-pair certificate). *For the official Version 2.3 Level-I $q = 19$, $(v, o, \ell, r) = (28, 5, 4, 4)$ public constants, all six maps E_{st} have rank 80. The full 80×680 symmetry-reduced feature map, the span of all self features, and the span of all cross features also each have rank 80. The linear self-feature output map E_{11}^{self} has rank 50.*

Author-reported exact computation. Each matrix is reconstructed from the official public constants and row-reduced by exact Gaussian elimination over $\mathbb{F}_{19}$; no floating-point rank test is used. The authors' audit scripts report the indicated pivot counts for all six cross maps and for the combined, self, cross, and E_{11}^{self} matrices. $\square$

Fix x_0, x_2, x_3 . After absorbing every verifier contribution independent of x_1 into $d \in \mathbb{F}_{19}^{80}$, all remaining cross terms are linear in x_1 . Let

$$C_{x_0, x_2, x_3} : \mathbb{F}_{19}^{132} \longrightarrow \mathbb{F}_{19}^{80}$$

denote their sum. The verifier equations restricted to these fixed columns have the exact form

$$C_{x_0, x_2, x_3}(x_1) + Q_1(x_1) = d. \tag{9}$$

For the clean two-column restriction $x_2 = x_3 = 0$ this specializes to

$$C_{x_0, 0, 0}(x_1) = E_{01} G_{01}(x_0, x_1) =: C_{x_0}(x_1).$$

The self term must be retained: it is quadratic in x_1 and takes values in the fixed 50-dimensional linear subspace $\mathcal{S}_1 = \text{im}(E_{11}^{\text{self}})$ of the output-coordinate space.

Corollary 10 (A sound 50-in-52 affine slice). *Let $C = C_{x_0, x_2, x_3}$ and assume its publicly computable coefficient matrix has rank 80. Choose any $a \in \mathbb{F}_{19}^{132}$ satisfying $C(a) = d$, and let $J : \mathbb{F}_{19}^{52} \rightarrow \ker(C)$ be a linear isomorphism. Then the solutions of Equation (9) on the affine slice $a + \ker(C)$ are in bijection with the solutions of*

$$Q_1(a + Jw) = 0, \quad w \in \mathbb{F}_{19}^{52}. \quad (10)$$

After choosing coordinates on $\mathcal{S}_1$, this is a system of 50 equations of degree at most two in 52 variables over $\mathbb{F}_{19}$.

Proof. Rank 80 makes C surjective, so a exists and $\dim \ker(C) = 132 - 80 = 52$. For $x_1 = a + Jw$, $C(x_1) = d$, and Equation (9) is therefore equivalent to $Q_1(a + Jw) = 0$. By Proposition 9, all values of Q_1 lie in the 50-dimensional linear subspace $\mathcal{S}_1$, so any linear coordinate isomorphism $\mathcal{S}_1 \cong \mathbb{F}_{19}^{50}$ yields 50 scalar equations. The affine parametrization is bijective on the selected slice, proving soundness and the claimed correspondence. $\square$

The rank condition on C is not universal. For the clean two-column slice, $x_0 = 0$ gives rank zero. The rank is publicly testable; on the audited key, all 100 sampled choices of x_0 gave $\text{rank}(C_{x_0}) = 80$. The attacker can retain any passing choice and solve the linear system $C(a) = d$. Fixed columns may also be chosen to address the verifier’s symmetric-block rejection, but then the rank of the resulting summed cross map must be tested anew; the clean C_{x_0} certificate alone does not prove this joint condition. The resulting slice is sound but not complete: it contains only the points in one target-matching coset of the kernel. The 50 equations in Equation (10) are generally inhomogeneous after the shift by a . Underdetermination by two variables does not prove existence. Under the stronger independent random-function heuristic, the expected number of roots is 19^2 and the zero-root probability is $(1 - 19^{-50})^{19^{52}} \approx e^{-19^2}$. Random quadratic systems need not satisfy that independence assumption, so this remains only a solvability heuristic for the target-matching slice.

The normal form is useful for both cryptanalysis and auditing because it exposes all remaining nonlinearity in a nearly square system. It does not by itself lower the conservative headline exponent: the Hashimoto estimator for the original 50-in-102 system already specializes almost all surplus variables before its square solve. We therefore tested whether the restricted 50-dimensional quadratic space has further public structure. Across 100 random restricted systems, the common polar radical was always zero, the quadratic span always had dimension 50, and every individual form had rank 51 or 52. On five independently generated keys, the adjoint/commutant algebra had dimension one. Random sampling of 10^5 linear combinations in the analogous Version 2.4 50-in-44 system found minimum rank 42, consistent with a generic high-rank matrix space. These checks provide negative evidence against an immediate radical, decomposition, or low-rank shortcut, while leaving specialized multigraded solving as an open route.

8 Concrete Complexity Estimates

8.1 Methodology

We evaluate the new residual systems with the methodology already used in the SNOVA submission and in Beullens’s paper. For a determined or overdetermined MQ system, the model estimates sparse Wiedemann/FXL from a semi-regular operating degree. Hashimoto’s method reduces an underdetermined system of m equations in n variables to a collection of smaller determined systems, optimizing over its parameters (a, k) [5, 1].

The Version 2.3 model evaluates an R_{old} -equation residual in $n\ell - M + R_{\text{old}}$ variables, up to its field-to-gate conversion. Its ordered-pair analysis takes R_{old} from an $m_1\ell^2$ -dimensional feature

Table 2: Dominant determined cores selected by the audited estimator snapshot under the stricter $p_1 = 1$ convention. The degree is the semi-regular operating degree used in the cost calculation. Degrees marked † are Hashimoto blocks that were not produced as explicit systems; they are the predicted operating degree only, not F4-audited (Section 9).

Level	Family	Public residual	Method	Dominant core	Degree
I	$\ell = 4$	50/102 or 50/98	square specialization	50/41	14
I	$\ell = 2$	48/112	Hashimoto $(a, k) = (2, 11)$	46/35	11 †
III	$\ell = 4$	70/146 or 70/142	square specialization	70/55	16
III	$\ell = 2$	72/168	Hashimoto $(a, k) = (2, 15)$	70/55	16 †
V	$\ell = 4$	90/182 or 90/178	square specialization	90/73	21
V	$\ell = 2$	96/224	Hashimoto $(a, k) = (2, 23)$	94/71	18 †

space. In the full-rank symmetry case we instead evaluate

$$\text{Hashimoto}(n\ell - M + K, K, q),$$

with $K = m_1 \binom{\ell+1}{2}$ as in Equation (8).

The estimator admits two conventions for the semi-regular series. In the analysis snapshot audited for this manuscript, the values in Version 2.3 Table 11 are numerically reproduced by using no extra homogenizing variable ($p_1 = 0$). Inserting one extra variable ($p_1 = 1$) increases the estimates by about 4.25 bits in most rows. We adopt the stricter, author-selected $p_1 = 1$ sensitivity convention in the headline table—so our figures are conservative relative to those $p_1 = 0$ numbers—and give both conventions in Table 3. The exact repository commit and estimator file must be fixed in the public artifact; without that pin, “current script” is not a stable provenance claim.

8.2 The determined systems actually costed by the estimator

The estimator does not run a Gröbner-basis algorithm directly on the full underdetermined residual systems displayed in our parameter tables. It first specializes surplus variables and then applies HybridMQ or Hashimoto’s reduction to smaller determined or overdetermined cores. Under the stricter $p_1 = 1$ convention, the load-bearing systems are summarized in Table 2.

For example, the strict Level-I $\ell = 4$ estimate first specializes the 50/102 system to a square 50/50 system and then guesses nine more variables, leaving the 50/41 overdetermined core whose modeled degree is 14. A solving-degree experiment on the untouched 50/102 presentation would therefore not directly validate the quoted work factor; the relevant experiment is on these estimator-selected cores.

Sensitivity to solving-degree error. The narrow Level-I $\ell = 4$ rows are genuinely sensitive to the semi-regular prediction. Raising the dominant degree from 14 to 15 multiplies the squared Macaulay dimension by approximately $((\binom{56}{15})/(\binom{55}{14}))^2 = (56/15)^2$, or about $2^{3.8}$. Thus degree 15 would place the strict estimate close to, but still approximately below, the 2^{143} target; degree 16 would likely erase the claimed shortfall. These two rows should therefore be read as narrow estimated shortfalls.

The larger instances have substantially more tolerance. One additional degree in the Level-III and Level-V dominant cores costs only roughly four bits in the same monomial-count model, compared with margins of 22.71–44.05 bits. The Level-I $\ell = 2$ row has a 12.70-bit margin. Accordingly,

Table 3: Estimates under both semi-regular conventions, conditional on $\rho = K$ and $\lambda_R = M - K$ for the attack relation. A simple relation has certified full quotient rank on all nine shapes; simultaneous quotient and affine rank for the rejection-bypassing relation is directly certified only on the audited (28, 5, 4, 4) Level-I shape. The $p_1 = 1$ column is the stricter convention used in the headline figures. The $p_1 = 0$ column numerically matches Version 2.3 Table 11 and the audited analysis snapshot.

Level	Parameters	Residual MQ	$p_1 = 0$	$p_1 = 1$	Target
I	(28, 5, 4, 4)	50/102	134.69	138.94	143
I	(48, 16, 2, 2)	48/112	126.05	130.30	143
I	(28, 4, 4, 5)	50/98	134.69	138.94	143
III	(40, 7, 4, 4)	70/146	180.05	184.29	207
III	(72, 24, 2, 2)	72/168	181.89	184.29	207
III	(38, 5, 4, 5)	70/142	180.05	184.29	207
V	(50, 9, 4, 4)	90/182	223.70	227.95	272
V	(96, 32, 2, 2)	96/224	234.52	238.77	272
V	(52, 6, 4, 6)	90/178	223.70	227.95	272

moderate anti-regularity could change the two narrow Level-I $\ell = 4$ conclusions without changing the Level-III or Level-V category conclusions.

8.3 All odd-characteristic parameter sets

For the nine $q = 19$ parameter shapes in Version 2.3, Equation (8) gives the residual systems shown in Table 3. The category targets are the values used by the submission: 143, 207, and 272 bits for Levels I, III, and V.

The Level-I $\ell = 4$ rows have the narrowest margin: 4.06 bits under the stricter convention. The convention choice moves the estimate by about 4.25 bits, more than this margin, yet both conventions place these rows below target (134.69 and 138.94 bits against 143; Table 3), so the shortfall does not turn on that choice. We nonetheless describe the Level-I rows as *estimated below target*. The remaining rows have margins of 12.70 to 44.05 bits under the same convention.

8.4 Version 2.4 analysis preview

The public SNOVA analysis repository was updated with six parameter shapes labeled Version 2.4. As of 24 July 2026, the official implementation repository’s supporting documentation still listed Version 2.3 and no matching Version 2.4 specification or implementation package was public [16, 14]. The symmetry quotient is algebraic and does not depend on the numerical parameters. If the preview retains the symmetric public-matrix construction, the stricter $p_1 = 1$ convention gives Table 4.

The first Level-I margin is the same narrow 4.06-bit estimate as in Version 2.3, and the second is only 1.43 bits. Those two rows should be described as estimated shortfalls. The Level-III and Level-V margins are substantially larger. Reparameterizing within the same symmetric construction must therefore use $m_1(\binom{\ell+1}{2})$, not $m_1\ell^2$, as the homogeneous rank ceiling.

8.5 Interpretation of the estimates

All displayed residual dimensions use the simultaneous full-rank equalities $\rho = K$ and $\lambda_R = M - K$ for the concrete relation and offsets used by the attack. A simple relation attains the first equality on every Version 2.3 shape, but that does not certify the pair for a rejection-bypassing relation. The

Table 4: Symmetry-reduced estimates for the Version 2.4 analysis preview, conditional on retention of the symmetric construction and on $\rho = K$, $\lambda_R = M - K$. These are not definitive Round 3 parameters; the table evaluates the public analysis-repository shapes under the same $p_1 = 1$ convention used for Version 2.3.

Level	Parameters (v, o, ℓ, r)	Residual MQ	New	Target	Margin
I	(26, 5, 4, 4)	50/94	138.94	143	-4.06
I	(43, 17, 2, 2)	51/103	141.57	143	-1.43
III	(37, 7, 4, 4)	70/134	184.29	207	-22.71
III	(64, 25, 2, 2)	75/153	194.57	207	-12.43
V	(44, 6, 4, 6)	90/146	227.95	272	-44.05
V	(90, 33, 2, 2)	99/213	246.84	272	-25.16

pair is certified in the audited (28, 5, 4, 4) Level-I end-to-end instances and is an explicit genericity assumption elsewhere, including the Version 2.4 preview. The parameter tables therefore require four attack-specific conditions: the evaluated public construction satisfies the stated symmetry hypothesis; the same relation and offsets attain the quoted quotient rank; the associated affine system has the quoted rank and is target-consistent (automatic when $\lambda_R = M - \rho$); and the residual fiber is nonempty, with every guessing or specialization step preserving a solution at the success probability charged by the estimator. Conditional on those points, the gate estimates inherit three standard solver assumptions:

1. the residual systems behave close enough to semi-regular systems for the operating-degree calculation to be meaningful;
2. sparse linear algebra has the costs assumed by the submission’s model;
3. field operations are converted to gates using the same coarse formula as the existing analysis.

These solver assumptions are neither introduced nor proved by this work. They are the methodology under which Version 2.3 claims its security levels. Our conclusion is therefore comparative and conditional: once the homogeneous feature dimension is corrected and the stated rank case is used, the candidate’s own framework places the odd-characteristic sets below their categories.

The Cabarcas-Li-Verbel-Villanueva-Polanco solver exploits additional multigraded structure in SNOVA systems [3]. We do not include any further speedup from that method.

The scaled F4 anomaly reported in [Appendix D](#) concerns the multibase graph-extension presentation $B_{\text{sym}}(x_1, \dots, x_c) = w + Ky$, which introduces linearly mixed kernel variables. It invalidates an unqualified production-scale extrapolation for that stronger proposed solver. It is not an experiment on the directly eliminated one-base residual systems costed in [Tables 1](#) and [3](#), and therefore neither validates nor contradicts their semi-regular model. The two polynomial presentations are scheme-theoretically equivalent as solution varieties, but Gröbner behavior need not be invariant under such an extended graph formulation.

9 Implementation and Algebraic Validation

The statements in this section are author-reported exact computations. They become independently checkable only when the public ePrint is accompanied by an immutable artifact containing the identified keys, matrices or rank certificates, scripts, raw logs, software versions, commands, seeds, resource limits, and a complete hash manifest. The present `.tex` source alone does not supply that evidence. Subject to that artifact qualification, we performed seven complementary layers of validation.

Universal identity. A self-contained checker verifies [Lemma 2](#) over randomly sampled symmetric matrices and constructs the duplication map explicitly. For $(m_1, \ell) = (5, 4)$ the ordered feature dimension is 80, the unordered feature dimension is 50, and the duplication map has rank 50.

All-parameter quotient-rank certificates. For the simplest public relation ($R_0 = I$ and $R_j = 0$ for $j \geq 1$), the quotient matrices attain full column rank for all nine $q = 19$ Version 2.3 shapes:

(v, o, ℓ, r)	matrix size	rank
(28, 5, 4, 4)	80×50	50
(48, 16, 2, 2)	64×48	48
(28, 4, 4, 5)	80×50	50
(40, 7, 4, 4)	112×70	70
(72, 24, 2, 2)	96×72	72
(38, 5, 4, 5)	100×70	70
(50, 9, 4, 4)	144×90	90
(96, 32, 2, 2)	128×96	96
(52, 6, 4, 6)	144×90	90.

For the Level-I $\ell = 4$ constants, all $19^3 = 6,859$ scalar-coefficient column relations $(R_0, R_1, R_2, R_3) = (I, c_1 I, c_2 I, c_3 I)$ in the affine chart $R_0 = I$ (the standard normalization; this is one chart of $\mathbb{P}^3(\mathbb{F}_{19})$, not all 7,240 projective points) produced an 80×50 matrix of rank 50, and 10,000 sampled module-valued relations did the same. These certificates show that the universal quotient ceiling is attainable on every Version 2.3 shape, rather than extrapolated from Level I. They do not by themselves certify the quotient rank of the rejection-bypassing relation used for a selected-target attack. A quotient-rank defect would reduce the equation count, but its complete effect still depends on the associated affine rank as described in [Proposition 7](#).

Official-key reconstruction. An independent Python implementation of the relevant Version 2.3 key-generation routines reproduced the Level-I (28, 5, 19, 4) KAT public key byte-for-byte. This checks the field arithmetic, the fixed A, B, Q generation (including the documented coefficient repair), and the public-matrix expansion used in the attack harness.

End-to-end verifier equivalence. On the KAT-derived (28, 5, 4, 4) key and two further generated keys of the same shape, we constructed the affine-column system for one fixed public relation with the skew offsets of [Lemma 8](#). These are the instances in which the manuscript directly certifies both the quotient and affine ranks for a rejection-bypassing attack family. Every trial gave

$$\text{rank}(\overline{E}_R) = 50, \quad \dim \ker(\overline{E}_R^\top) = 30, \quad \text{rank}(W_R L_R) = 30.$$

After eliminating the 30 affine equations, a known solution satisfied the residual 50-equation system, and reconstructing the original 80 verification coordinates gave zero discrepancy. The same harness verified that the offset construction of [Lemma 8](#) made symmetric-block rejection impossible for every residual assignment.

Cross-column normal form and restricted structure. The complete Level-I symmetry-reduced feature map has size 80×680 and rank 80. Every individual cross-column submap is 80×80 and invertible, yielding [Proposition 9](#). For 100 sampled choices of the fixed column x_0 , the coefficient map C_{x_0} had rank 80 and a 52-dimensional kernel. Restriction of the 50 self-quadratics to

those kernels gave common-radical dimension zero in every trial, individual ranks 51 or 52, and a full 50-dimensional quadratic span. A separate adjoint-algebra computation on five keys returned only the scalar algebra. These experiments were added specifically to test whether the 50-in-52 normal form hides an easier decomposition; none was observed.

Low-degree sanity check. For the exported official 50-in-102 residual, the degree-one Macaulay multiplication matrix had full column rank (5,150 columns) and no unexpected degree-one syzygy. More substantially, the planted residual root has full Jacobian rank 50. Among 10,000 random coordinate 50-dimensional strips through that root, 9,466 had invertible restricted Jacobian, closely matching the random-matrix probability $\prod_{j=0}^{49} (1 - 19^{j-50}) = 0.9445987429 \dots$. Across 20 independently generated official-shaped keys, every planted root again had Jacobian rank 50, and the number of the 10,000 sampled coordinate strips retaining full rank ranged from 9,412 to 9,491 (mean 9,447).

All 102 coordinate directions, the planted-root direction, and 10,000 random nonzero directions produced polar matrices of full output rank 50. We also evaluated every projective Hessian combination whose coefficient vector has Hamming weight at most two, $50 + \binom{50}{2}(19 - 1) = 22,100$ combinations, together with 20,000 uniformly random nonzero combinations. The minimum observed Hessian rank was 100 in 102 variables. These tests provide strong negative evidence against a common radical, a low-rank pencil, or an obvious decomposition.

They do not by themselves prove that the production systems have the semi-regular operating degrees in Table 2. We therefore ran the direct solver audit described above: from the official KAT keys we specialized each residual to its estimator-selected $\ell = 4$ core and compared its degree-by-degree Faugère F4 behaviour (per-degree Macaulay matrix size, new basis elements, and zero reductions) against an ensemble of five random systems matched in field, dimensions, quadratic support density, and consistency status. Because a complete solve at the modeled degree is itself of the order of the estimated attack, no run completes; the reproducible quantity is the per-degree structure up to the maximum degree reached under a fixed memory/time budget.

For the load-bearing Level-I $\ell = 4$ core (50/41, modeled degree 14), the official system showed no qualitative deviation from the five matched random controls through the maximum degree reached (F4 degree 4 completed; degree 5 entered before the budget cut both official and controls): the degree schedule matched, no early degree fall appeared, no zero-reduction surplus beyond the F4 Koszul artifacts common to the controls was observed, and no Macaulay-rank deficit appeared. The official core lay within the control ensemble’s spread for the recorded quantities. This small, truncated comparison is consistent with semi-regular behaviour, but it is not a statistical indistinguishability test and does not approach the modeled degree 14 solve. The row therefore remains a narrow estimated shortfall. The Level-III and Level-V $\ell = 4$ cores (70/55 and 90/73) are larger still and lie even further beyond the completion budget than the 50/41 core, so no F4-audited degree is available for them and their rows stay modeled predictions.

Scaling ladder. Because the load-bearing cores are too large to solve to completion, we validate the operating-degree model on a ladder of *faithful* symmetry-reduced cores small enough to run msolve to a full rational parametrization. Each core is built through the same verified pipeline (residual construction, reduction, and specialization), passes the identity and planted-root checks, and is kept over $q = 19$ so that field equations (active only at degree 19) never contaminate the measured degree. For each core we record the maximum F4 round degree and compare against three random controls matched in dimensions, quadratic-support density, and planted-root consistency (Table 5).

Table 5: Observed maximum F4 round degree (msolve, run to completion) versus the semi-regular prediction on faithful reduced cores. K is the quotient-capped feature-map rank; each core is determined ($m = n = K$). Daggered entries sit one below the semi-regular first-fall index: on both, the F4 bulk table is identical to a matching rung that reaches the predicted degree (e.g. the two 12/12 cores share an identical degree schedule and peak Macaulay size), and the ± 1 is a trailing spair-reduction label in msolve, not a degree fall. No audited core reaches a degree above its prediction. The three controls per rung are a small qualitative comparison, not a formal statistical test.

ℓ	K (m/n)	density	predicted D_{reg}	observed F4 degree	random controls
2	3 (3/3)	0.72	4	4	3, 4, 3
2	6 (6/6)	0.94	7	7	7, 7, 7
2	9 (9/9)	0.95	10	10	5, 10, 10
2	12 (12/12)	0.96	13	12 [†]	7, 12, 12
3	6 (6/6)	0.90	7	7	4, 6, 4
3	12 (12/12)	0.95	13	13	7, 7, 13
4	10 (10/10)	0.96	11	10 [†]	6, 10, 11

On five of the seven rungs the observed solving degree equals the semi-regular prediction exactly; on the two daggered rungs it is one lower for the bookkeeping reason noted in the caption. In this ladder no audited core reaches a degree above its prediction. Relative to the three matched controls per rung, the official core lies within the observed control range on six rungs and one degree above it on one ($\ell = 3$, $K = 6$: 7 versus a maximum of 6). This small sample is qualitatively consistent with the operating-degree model, but it is too small for a claim of statistical indistinguishability or a reliable estimate of run-to-run variation. It gives direct F4 corroboration on genuinely reduced SNOVA cores only up to $K = 12$; extrapolation to the headline dimensions continues to rest on the semi-regular predictor.

The three $\ell = 2$ rows of Table 2 are the dominant determined *blocks* of a Hashimoto change-of-basis reduction; producing those exact blocks as explicit polynomial systems is outside the current reproducibility tooling, so their operating degrees remain the modeled semi-regular predictions (agreed by two independent predictors) and were *not* F4-audited. As a structural cross-check on the same official $\ell = 2$ keys we instead audited the valid but not gate-optimal square-specialization cores (48/36, 72/60, 96/79), which are coordinate-emittable and verified byte-for-byte against the reference public key. They should be included in the immutable reproducibility artifact described in Appendix E. These cores are themselves above the completion budget ($n \geq 36$), so they are not F4-audited; the direct degree corroboration comes from the smaller faithful cores of Table 5.

10 Countermeasures and Standardization Implications

The immediate requirement is to evaluate every symmetric-matrix parameter set using

$$K_d = m_1 \binom{d+1}{2}, \quad d = \dim_{\mathbb{F}_q} \mathbb{F}_q[S],$$

as the maximum homogeneous rank in the affine-column attack; the current irreducible construction has $d = \ell$. Merely choosing A, B, Q matrices for which the ordered-coordinate E_R has high rank cannot overcome the quotient.

Possible design responses include:

1. **Reparameterization.** Increase the dimensions until the symmetry-reduced residual MQ system meets the desired category under the team’s accepted cost methodology. Under the rank and solver assumptions used in the tables, all three NIST categories fall short, and the Level-III and Level-V shortfalls of up to 44 bits demand larger parameters than the narrow Level-I margin alone would suggest, working against the compact keys that motivated the odd-characteristic design.
2. **Remove public-matrix symmetry.** This sacrifices the key-size reduction that motivated the odd-characteristic change, but removes [Lemma 2](#) in its present form.
3. **Redesign the feature layer.** Preserving ordered power-pair coordinates requires changing the public forms or the left/right power actions so that swapping the two labels no longer gives the same quadratic polynomial. Any such construction should then prove rank on its actual feature space. Merely adding or randomizing more A, B, Q emulsifier terms inside the present symmetric construction cannot restore the discarded alternating directions.
4. **Publish joint rank certificates.** For every proposed parameter set, evaluate both the quotient rank ρ and the affine rank λ_R on the concrete public construction and on the relations used by the attack. A quotient-rank certificate alone does not establish selected-target solvability; an affine-rank defect must be charged through the corresponding target-consistency or resampling factor.

The symmetric-block rejection should not be counted as a countermeasure against the affine-column attack, because fixed offsets can satisfy it identically. A rejection rule capable of blocking the attack would need to constrain the affine family itself without destroying correct signing, rather than test a property that the attacker can set with constants.

NIST advanced SNOVA to Round 3 while allowing updated specifications and implementations [11]. The Version 2.3 draft and Version 2.4 analysis preview are therefore at an appropriate stage for correcting the attack model. Returning unchanged to the submitted $q = 16$ construction is not an obvious remedy, because the structure-aware wedge attack already reports those parameters broken. Any viable repair must address both attack families and be analyzed on the actual, symmetry-reduced feature space.

11 Conclusion

The public odd-characteristic SNOVA redesign makes its scalar-expanded base matrices symmetric. That key-size optimization identifies the ordered power-pair coordinates used by the Beullens affine-column attack. The homogeneous feature map factors through

$$\bigoplus_{i=1}^{m_1} \text{Sym}_{\mathbb{F}_q}^2(\mathbb{F}_q[S])$$

and has rank at most $m_1 \binom{\ell+1}{2}$, not $m_1 \ell^2$. This is a universal algebraic identity, not a weak-key phenomenon.

Under the candidate’s own semi-regular Hashimoto/Wiedemann framework, and in the full-rank case $\rho = K$, $\lambda_R = M - K$, every $q = 19$ Version 2.3 set and every Version 2.4 preview set is estimated below its intended category. For a simple relation, full quotient rank is certified for all nine Version 2.3 shapes. Simultaneous full quotient and affine rank for the rejection-bypassing relation is certified end-to-end on the audited (28, 5, 4, 4) Level-I shape and remains an explicit genericity assumption for the other shapes. The Level-III and Level-V shortfalls are robust within a substantially wider range of solving-degree error. The Level-I $\ell = 2$ row retains a 12.70-bit estimated

margin. The two Level-I $\ell = 4$ rows are narrower: the strict estimator costs actual 50/41 cores at modeled degree 14, and a rise to degree 16 could erase their 4.06-bit shortfall. We therefore describe those rows specifically as narrow conditional estimates rather than placing all nine rows on an equal empirical footing.

The verifier’s symmetric-block rejection is bypassed constructively by fixed skew offsets. Independently, when a sampled fixed column gives a surjective cross map, choosing a target-matching kernel coset restricts the official Level-I verifier fiber to a sound 50-equation system on a 52-dimensional affine slice. The official Level-I residual exhibits full planted-root Jacobian rank, full sampled polar rank, and very high sampled Hessian rank. These facts rule out several obvious structural degeneracies without proving the semi-regular operating degree.

The submitted $q = 16$ package is outside this exact quotient because its public matrices are not symmetrized, but it is already heavily affected by wedge attacks. The present result therefore targets the public odd-characteristic repair direction rather than offering a safe fallback. What remains uncertain is not only solver cost: outside the audited Level-I instances the simultaneous affine-rank and target-consistency conditions are unproved; the clean cross-column surjectivity certificate has not been shown to coexist with the fixed offsets used for the rejection bypass; the load-bearing $\ell = 2$ Hashimoto blocks were not emitted and F4-audited as explicit systems; and the computational certificates are not independently reproducible until an immutable artifact is archived. The symmetry quotient and rank ceiling are universal. The selected-target reduction to the displayed dimensions is exact whenever the measured equalities $\rho = K$ and $\lambda_R = M - K$ hold. The cross-column normal form is an exact, sound restriction conditional on its public surjectivity test, but not a complete parametrization of the verifier fiber. The displayed gate exponents additionally inherit the heuristic semi-regular methodology used by the SNOVA submission.

A Parameter Formulas

For convenience, Table 6 records the intermediate dimensions. Recall

$$n = v + o, \quad m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \quad M = or\ell, \quad K = m_1 \binom{\ell + 1}{2}, \quad N_{\text{res}} = n\ell - M + K.$$

Table 6: Full-rank residual dimension calculation for all $q = 19$ Version 2.3 draft sets ($\rho = K$, $\lambda_R = M - K$).

Level	(v, o, ℓ, r)	m_1	M	K	N_{res}
I	(28, 5, 4, 4)	5	80	50	102
I	(48, 16, 2, 2)	16	64	48	112
I	(28, 4, 4, 5)	5	80	50	98
III	(40, 7, 4, 4)	7	112	70	146
III	(72, 24, 2, 2)	24	96	72	168
III	(38, 5, 4, 5)	7	100	70	142
V	(50, 9, 4, 4)	9	144	90	182
V	(96, 32, 2, 2)	32	128	96	224
V	(52, 6, 4, 6)	9	144	90	178

B A Linear-Algebra View of the Quotient

For one base form, order the coordinates of $\mathbb{F}_q^{\ell^2}$ by pairs (a, b) . Order the coordinates of $\text{Sym}^2(\mathbb{F}_q^\ell)$ by pairs $a \leq b$. The duplication matrix D_ℓ has one 1 in row (a, a) and, for $a < b$, one 1 in each of rows (a, b) and (b, a) . It has full column rank $\binom{\ell+1}{2}$ over every field. For m_1 base forms,

$$D = I_{m_1} \otimes D_\ell.$$

The attack matrix used in the odd-characteristic homogeneous system is $E_R D$. In particular, even an ordered-coordinate matrix E_R of full row and column rank cannot produce rank above $\text{rank}(D) = K$.

For $\ell = 4$ the single-form duplication matrix maps ten coordinates to sixteen positions. Its six-dimensional complementary alternating subspace is spanned by the differences $e_{a,b} - e_{b,a}$ for $a < b$ (using the odd-characteristic direct-sum identification). These six directions are always killed; additional kernel may occur after composition with the public output map. Across $m_1 = o$ square instances, the universal quotient removes $6o$ ordered coordinates. For the Level-I $o = 5$ set, the audited relation has full quotient rank and hence a 30-dimensional left kernel, exactly matching the affine-linear block observed end-to-end.

C Why the Extension Field Does Not Give an Immediate MinRank Break

The field notation $A = \mathbb{F}_q[S] \cong \mathbb{F}_{q^\ell}$ makes a tempting but false inference: one might try to regard the public matrices as an A -linear matrix code of size $n \times n$ and search for an A -rank-one dual element. The structural proof above does not justify that step. For fixed u , write $\Phi_{u,i}(\xi, \eta) = \Psi_i(\xi \otimes \eta)(u)$. This pairing is symmetric and bilinear over $\mathbb{F}_q$, but it is generally not A -balanced:

$$\Phi_{u,i}(\alpha\xi, \eta) \neq \Phi_{u,i}(\xi, \alpha\eta).$$

Such balance would require the public form to commute appropriately with the A -action. In SNOVA, the secret transform and the Q factors lie in A , but the base quadratic blocks are sampled as general matrices subject only to the UOV and, in odd characteristic, global symmetry conditions. The specification itself treats the extension-field decomposition as a collection of Frobenius-conjugate lifted UOV instances, not as one A -linear public matrix code.

An official-KAT audit of the Level-I $(28, 5, 19, 4)$ key gives a particularly sharp falsification. With $J = I_n \otimes S$, all five public matrices satisfy

$$\text{rank}(P_i J - J P_i) = 132 = n\ell.$$

None of their $5 \cdot 33^2 = 5,445$ scalar 4×4 blocks lies in the centralizer field A . The two-sided power orbit has the full tensor-square dimension $5\ell^2 = 80$, while its symmetric quotient has the full dimension $5\binom{\ell+1}{2} = 50$; it does not collapse further to the $5\ell = 20$ dimensions that an A -balanced model would predict. Finally, a genuine public oil dual $uu^\top$ has $\mathbb{F}_q$ -rank one, fails to commute with J , and has nonzero blocks of rank one. A nonzero multiplication matrix from the field A is invertible, so this dual is not an A -valued matrix.

Hence there is no justified $n \times n$ rank-one MinRank problem over $\mathbb{F}_{q^\ell}$ that replaces the scalar-expanded problem. The valid surviving uses of the extension field are the known Frobenius/diagonal-block lifting attacks and the A -valued Gram relations analyzed above.

D The Multibase Stable-Ideal Extension

[Proposition 5](#) gives an exact attack architecture beyond the one-base reduction. Specializing to the current irreducible degree- ℓ construction, for a c -base relation let

$$F_c = m_1 \binom{c\ell + 1}{2}, \quad M = or\ell, \quad p_c = F_c - M.$$

When $F_c \geq M$ and the public feature map is surjective, select a preimage w of the target and a basis K of its kernel. Forgery becomes

$$B_{\text{sym}}(x_1, \dots, x_c) = w + Ky, \tag{11}$$

with p_c linear kernel variables. For $c = r$, this is a symmetry-aware stable encoding of direct forgery. Surjectivity makes every target lie in a linear output fibre, so no separate *linear* membership search is needed; it does not imply that the nonlinear Gram map has a preimage in that fibre.

After diagonalizing the public A -action, there remain ℓ grading blocks, not $c\ell$ blocks. If each block contains $s = M/\ell = or$ specialized quadratic variables, the numbers of feature equations of each multidegree are

$$A_c = m_1 \binom{c+1}{2} \text{ of degree } 2e_a, \quad B_c = m_1 c^2 \text{ of degree } e_a + e_b.$$

The corresponding formal complete-intersection series is

$$H(\mathbf{t}) = \frac{\prod_a (1 - t_a^2)^{A_c} \prod_{a < b} (1 - t_a t_b)^{B_c}}{\prod_a (1 - t_a)^s}.$$

Using the Cabarcas-style nonhomogeneous criterion gives attractive formal screens. For the Level-I square $\ell = 4$ set, $c = 4$ gives about $2^{99.8}$ operations with an $\omega = 2$ sparse-linear-algebra model and $2^{132.3}$ with $\omega = 2.81$. These figures are *not* part of the headline attack claim. They depend on a solving-degree extrapolation that has not been validated at production scale.

Scaled experiments reveal the risk. Generic lifted systems with $(\ell, c, m_1) = (4, 4, 1)$ reached maximum F4 degrees 4 and 6 when the number of variables per grading block was one and two, respectively; the next scale already produced very large degree-three matrices before completion. A separate $(3, 3, 1)$ family reached degree 6 where the formal criterion predicted roughly 3–4. These are ordinary-F4 experiments rather than a custom multigraded implementation, so they neither prove nor refute the attack estimate. They do show that a public complexity claim near 2^{100} would be premature without an independent multigraded-F4 or Magma validation.

The graph extension is scheme-theoretically equivalent to direct forgery: projection onto the x -coordinates identifies its solution variety with the corresponding fiber of the direct feature map. It should not, however, be confused with the directly eliminated one-base polynomial presentation. The public kernel generally mixes the nominal grading sectors, so the graph-extension variables do not decompose into the independent homogeneous blocks assumed by the most optimistic formal complete-intersection screen. Gröbner behavior can therefore differ substantially even though the two presentations encode the same solutions.

The exact conclusions are therefore: the multibase factorization, feature counts, graph equivalence, and surjectivity tests are valid; the scaled F4 anomaly applies to this extended formulation; and the stronger multibase work factors remain an unconfirmed solver extrapolation. It supplies no direct evidence for or against the one-base estimator cores in [Table 2](#).

E Reproducibility and Claim Checklist

A public claim based on this work should distinguish the following statements.

Proved algebraically. The homogeneous affine-column system factors through unordered power pairs and has rank at most $m_1(\binom{d+1}{2})$, with $d = \dim \mathbb{F}_q[S]$; the verifier’s symmetric-block rejection can be bypassed by explicitly computable fixed skew offsets; and [Theorem 6](#) is an exact solution-preserving reduction conditional on target consistency.

Author-reported computations. The manuscript reports that the Level-I official-key reconstruction matches a KAT public key; one rejection-bypassing relation on the KAT-derived $(28, 5, 4, 4)$ key and two generated keys of that shape has $\rho = 50$ and $\lambda_R = 30$ and yields an exact 50-in-102 system; all six official Level-I cross-feature output maps are invertible, and 100 sampled fixed columns pass the surjectivity test needed for the sound 50-in-52 slice; a separate simple relation has full quotient-rank certificates on all nine Version 2.3 shapes; and the official Level-I residual has full sampled polar rank, full planted-root Jacobian rank, and high-rank sampled Hessian combinations. These become independently certified only through the immutable artifact specified below.

Assumed for untested shapes. The all-shape selected-target dimensions require the same attack relation and offsets to satisfy $\rho = K$ and $\lambda_R = M - K$. These simultaneous equalities are directly certified only on the audited $(28, 5, 4, 4)$ Level-I shape; a defect elsewhere would alter the residual dimensions or require the target-consistency factor in [Proposition 7](#).

Estimated heuristically. The gate complexities in [Tables 1](#) and [3](#), like the submission’s estimates, assume that the sampled residual fiber is solvable with the modeled probability, semi-regular behavior, the modeled success of guessing and specialization, and a particular sparse-linear-algebra model. The two Level-I $\ell = 4$ rows are sensitive to a one- or two-degree error; the Level-III and Level-V margins tolerate several additional degrees in the same model. The load-bearing $\ell = 2$ Hashimoto blocks are prediction-only and were not explicitly generated or F4-audited.

Audited negative result. The public forms are not A -linear and there is no immediate $n \times n$ MinRank shortcut over $\mathbb{F}_{q^\ell}$.

Preliminary extension. The exact multibase Gram attack is included as a solver-research direction, but its stronger formal exponents are not claimed as validated production work factors.

Not claimed. We do not claim a practical production-size forgery, a key-recovery attack, a break of the $q = 16$ parameter sets, a universal full-affine-rank theorem, a proof that every restricted target fiber is nonempty, or a proven running-time upper bound matching the estimates.

The public ePrint should archive an immutable reproducibility artifact at the same time as the manuscript. At minimum, that archive must contain:

- exact repository commits, local patches, parameter files, KAT identifiers, and generated public-key digests;
- a self-contained checker for the symmetry identity, quotient map, parameter formulas, and rejection bypass;
- the concrete matrices or machine-verifiable rank certificates for every quotient-rank, affine-rank, and cross-map claim, together with a verifier;
- a pinned implementation of both semi-regular conventions, the Hashimoto optimization, the multibase Hilbert-series calculation, and all exact cost tables;
- rank-search, Jacobian, coordinate-strip, polar, Hessian, and degree-one Macaulay scripts and raw logs;
- the solving-degree experiments, including emitted polynomial systems, matched controls, solver versions, commands, seeds, resource limits, and complete per-degree F4 logs;
- end-to-end KAT and verifier-equivalence transcripts; and
- a manifest giving the SHA-256 digest of every source, input, output, and evidence file, plus a

documented one-command reproduction path. Until such an archive is linked, the computational statements above should be read as author-reported rather than independently reproduced. A hash of the specification PDF alone is useful but insufficient to identify the code, keys, certificates, logs, and local modifications. The SHA-256 digest of the Version 2.3 PDF used in the implementation audit is

963fa43d7e6ed8ecf1e77556e7cf47f62d4c34177efece2cd9a9517d0fbaf783.

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